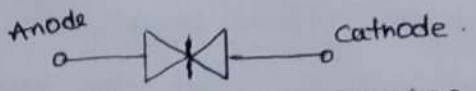
① Explain how negative resistance is created in Gunn diodes using two valley theory?

A Gunn Diode:-

It is a type of diode having two terminals which allows the current to flow in both directions with negative resistance used for its very high switching speed.

Symbol:-

* It is consists of two conventional diodes connected face to face.



* It is also known as "Transfer Electron Device (TED)"

Construction:-

⇒ It is made of only an N-type semiconductor.

⇒ It consists of 3-layers of N-type semiconductor.

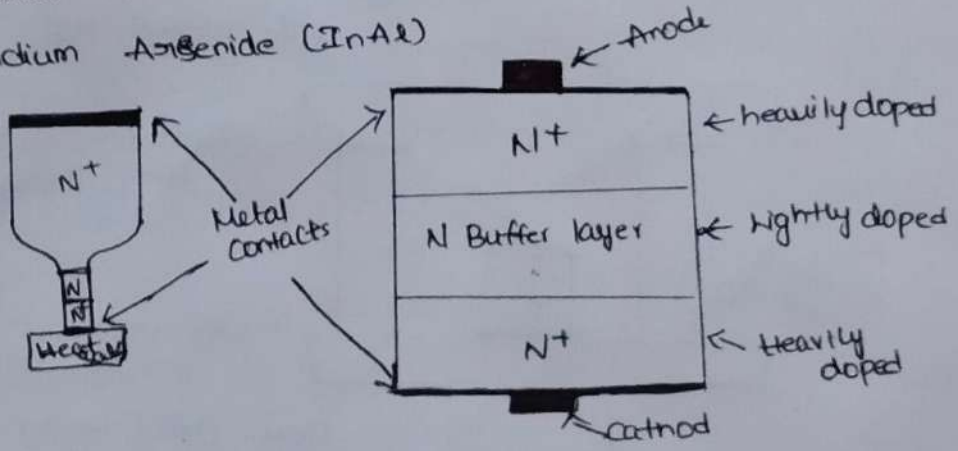
⇒ The materials used for Gunn diode are,

Gallium Arsenide (GaAl)

Gallium Nitride (GaN)

Indium phosphide (InP)

Indium Arsenide (InAl)



⇒ The two Extreme End layers are heavily doped.

⇒ While the middle layer (thin layer) is lightly doped an known as buffer layer.

⇒ The width of the middle layer determines the operating frequency of the device.

⇒ The metalic contact for used for providing biasing (input, to the diode).

⇒ Heat sink is connected to prevent over heating of diode.

Working Principle:-

⇒ When voltage is applied an Electric field is Exhibited across the active layer (middle layer).

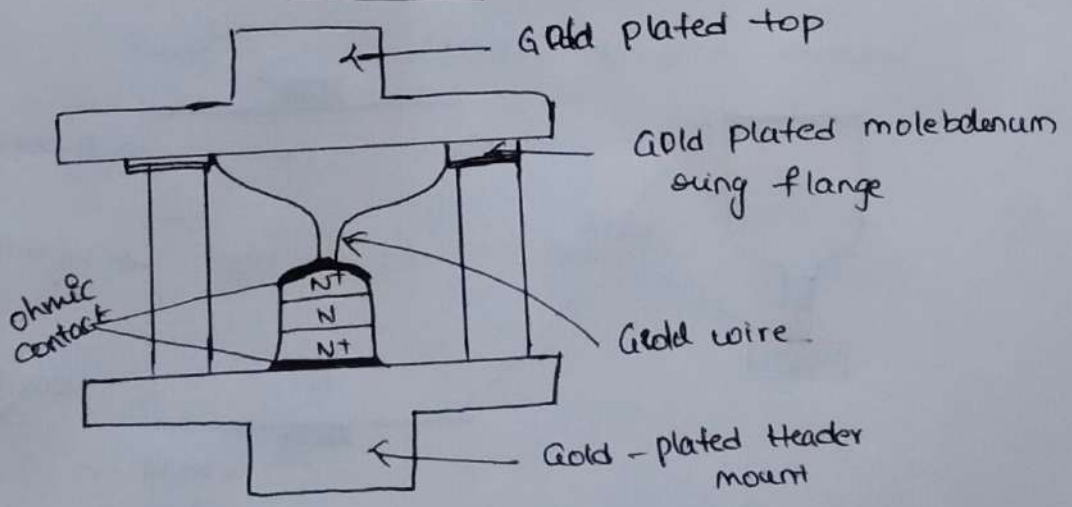
⇒ The current flow increases with the increases in applied voltage.

⇒ Hence the device has +ve resistance.

⇒ The current increase until the voltage increases upto the threshold voltage "Vth".

⇒ When the voltage increases beyond the "Vth". threshold voltage the current density across the active layer decreases.

⇒ This is known as "Gunn Effect"

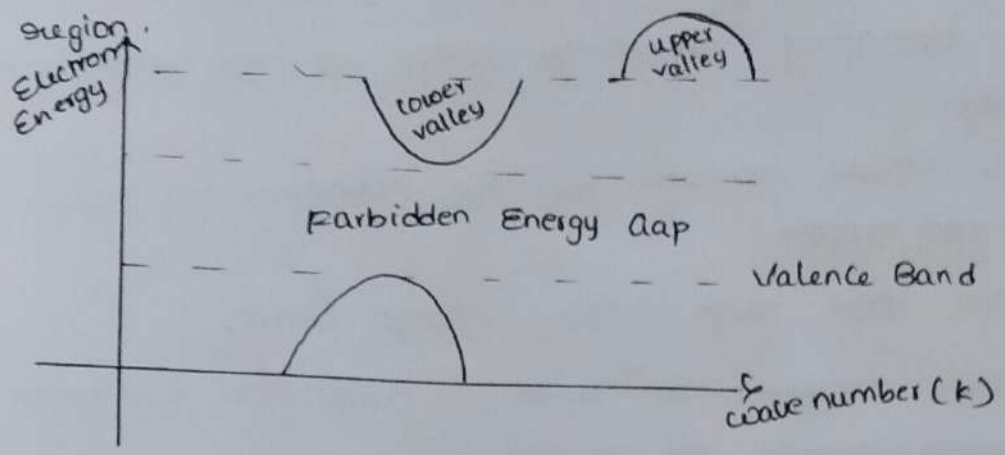


⇒ The current increases with increase in applied voltage

⇒ In this region the device exhibits "negative resistance"

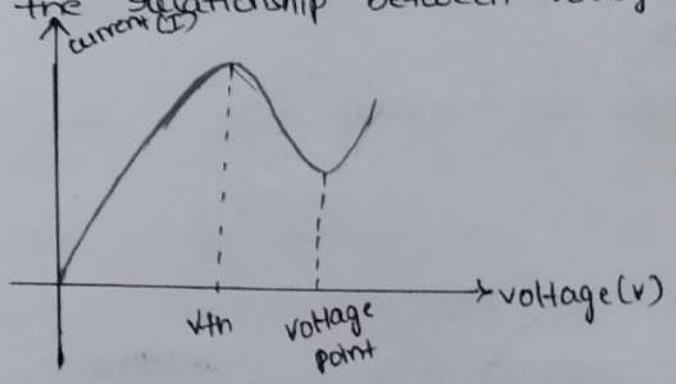
⇒ This doesn't allow the another current pulse to pass through it.

⇒ Hence the operating frequency depends upon the length of the active region.



VI characteristics:-

⇒ It shows the relationship between voltage & current through Gunn diode.



Advantage:-

⇒ It offers very high operating frequency → 10GHz to 200GHz

⇒ It is of small size & occupies little space.

⇒ It has a large operating bandwidth.

Disadvantages:-

⇒ It operates at Negative resistance region at high voltage

⇒ It has low stability & high Noise figure.

Applications:-

⇒ It is used in microwave TX & RX

⇒ used in Radio communication devices & sensors.

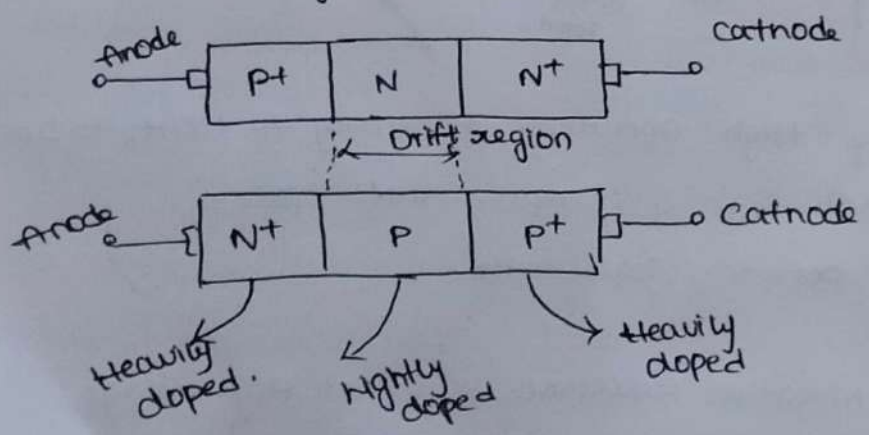
(8) Describe the principle of operation of TRAPATT Diode with suitable diagram.

=> TRAPATT Diode:-

* TRAPATT :- Trapped plasma Avalanche triggered ~~transalt~~ transit time diode.
* It is a microwave frequency generator
* The operating frequency range of the diode is from 3GHz-100GHz.
* This diode operates by the creation of plasma in the depletion region.
* plasma has high density charge carries.
* TRAPATT diodes can used as amplifiers oscillators at a frequency range of 0.15GHz to 100GHz.
* The Efficiency of TRAPATT diodes is more compared to the IMPATT diode.
* The construction of TRAPATT diode is simple & Easy to design.

Construction:-

=> It is a 3-layers structure.



⇒ The Extreme Ends of the diode are Heavily doped.

⇒ The middle layer is lightly doped.

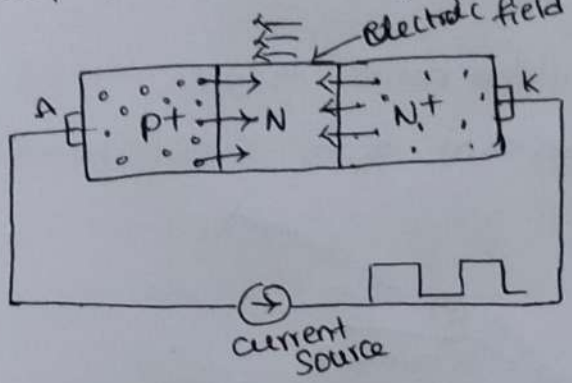
⇒ The middle layer region is called as the drift region.

⇒ The metal contact are used for providing / applying the biasing voltage.

⇒ The two terminals of the diode are anode & cathode.

Working Principle:-

⇒ The TRAPATT is a 3-layer structure diode.



⇒ Initially the diode is connected in reverse biased condition

⇒ The current source is connected to the cathode by which the current passes from N+ to p+ region.

⇒ Now the majority charge carries from N+ region tends to move towards p+ region & vice-versa.

⇒ Due to this diode acts as a capacitor and charge gets increased.

⇒ The increasing points of charge are from A→B.

⇒ As more amount of current gets increased then the electric field in the depletion region gets increased.

⇒ Due to this the capacitor starts slowly discharging

⇒ The discharging points are from B to C.

⇒ As more & more amount of current gets increased the capacitor

discharge more & more.

⇒ So again the discharging points are from to C to D.

⇒ Hence in order to overcome this situation, slowly pull out the electrons & holes from the depletion region.

⇒ Due to this the capacitor changes it's state from discharge ~~the~~ to charging.

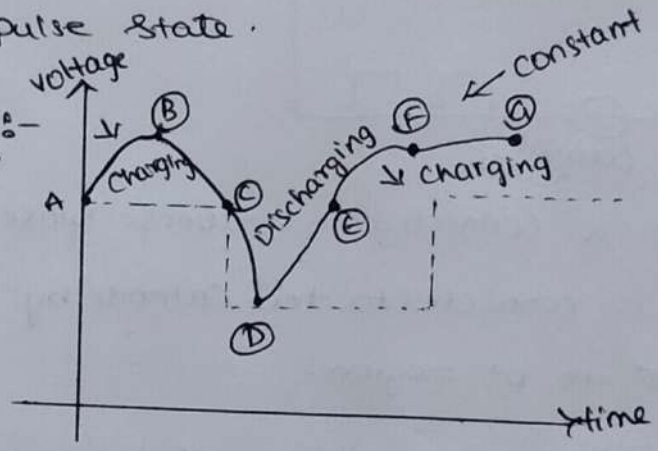
⇒ Now the charging points are from D to E

⇒ Further pulling of charge carries make the capacitor charge more & more.

⇒ So again it charge to the points E to F.

⇒ Finally ~~it's~~ remains constant from F to G points till the next clock pulse state.

Characteristics:-



Advantages:-

* High Efficiency
* Low power dissipation
* Good Frequency range
* High o/p power (1.2kW at 1.5GHz)

Disadvantage:-

* Extremely high noise Figure (60dB)
* unsuitable for continuous wave operation

Applications:-

* Radar System
* Navigational Systems
* Radio Equipments.

3. Which is the principle of operation of IMPATT Diode with Suitable diagram.

A) ⇒ IMPATT – Impact Ionization Avalanche Transit time diode.

⇒ It is a high frequency microwave diode.

⇒ The frequency range is from 1GHZ to 100GHz.

⇒ It provides high power handling Capability.

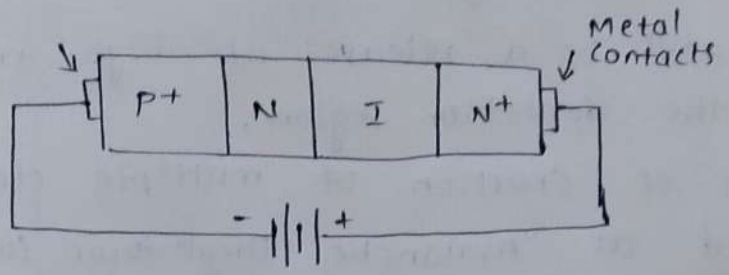
⇒ It is operated at "Negative Resistance region".

⇒ The applying of reverse bias to the diode make it operate at a breakdown voltage called as "Avalanche Breakdown voltage".

Construction :-

⇒ It is constructed by using 4 layers.

⇒ The layers are $P^+ - N - I - N^+$.



⇒ The IMPATT diode can be constructed using different materials like,

* Gallium Arsenide (GaAs)
* Silicon (Si), Germanium (Ge)
* Indium phosphate (InP).

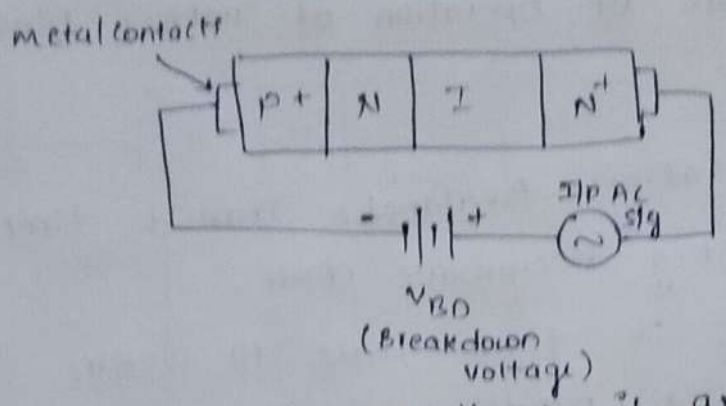
⇒ GaAs is mostly choosen because of low noise behaviour property.

Working principle :-

⇒ It works on mainly two important principles

Avalanche Breakdown

Transit time delay.



⇒ When reverse bias voltage is applied to the diode then the width of the depletion region becomes wider.

⇒ Now the minority charge carriers in both p-type & n-type semiconductor starts moving/crosses over each other.

⇒ Due to this there will be formation of "electron-hole pairs".

⇒ This pairs of electron-hole gets multiplied & more number gets to be generated.

⇒ Hence there will be a release of high amount of current over the depletion region.

⇒ This situation of creation of multiple electron-hole pairs is called as "Avalanche Breakdown (or) Impact Ionization".

Transit Time Delay :-

⇒ The time taken by the charge carriers which are moving from P-type to the N-type semiconductor device is termed as "Transit time delay".

Transit time delay,

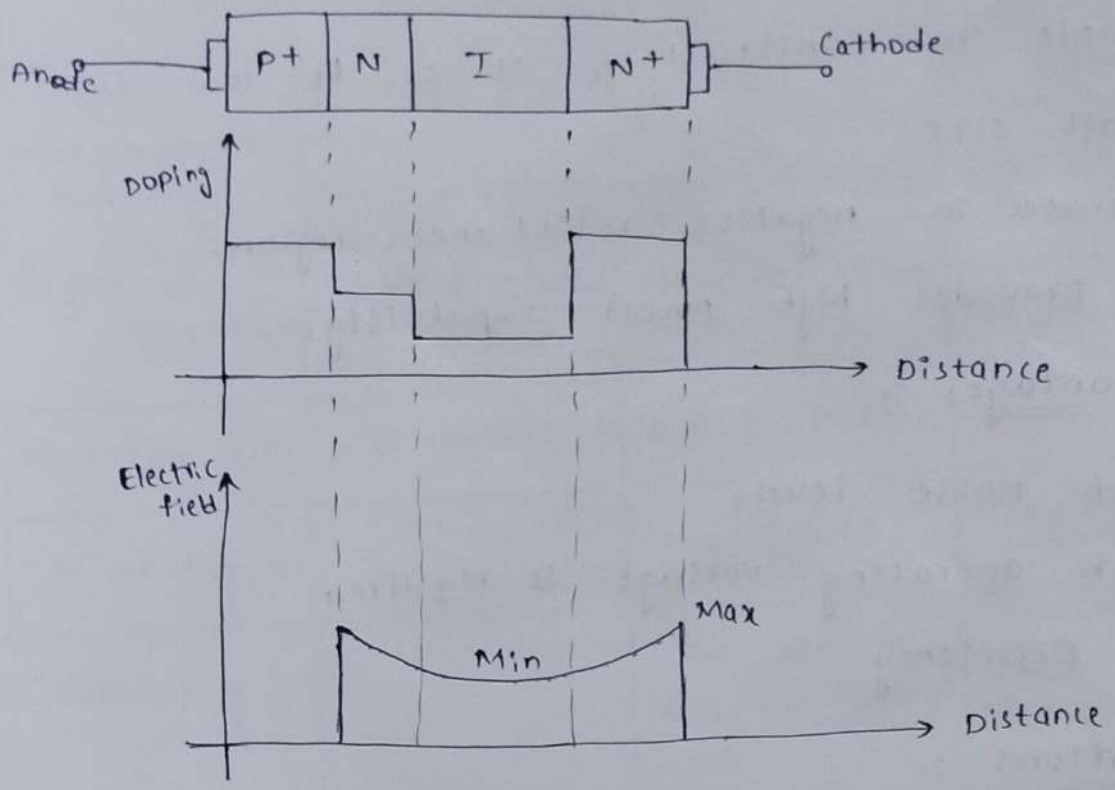
$$\boxed{\tau = L/v_s}$$

Where,

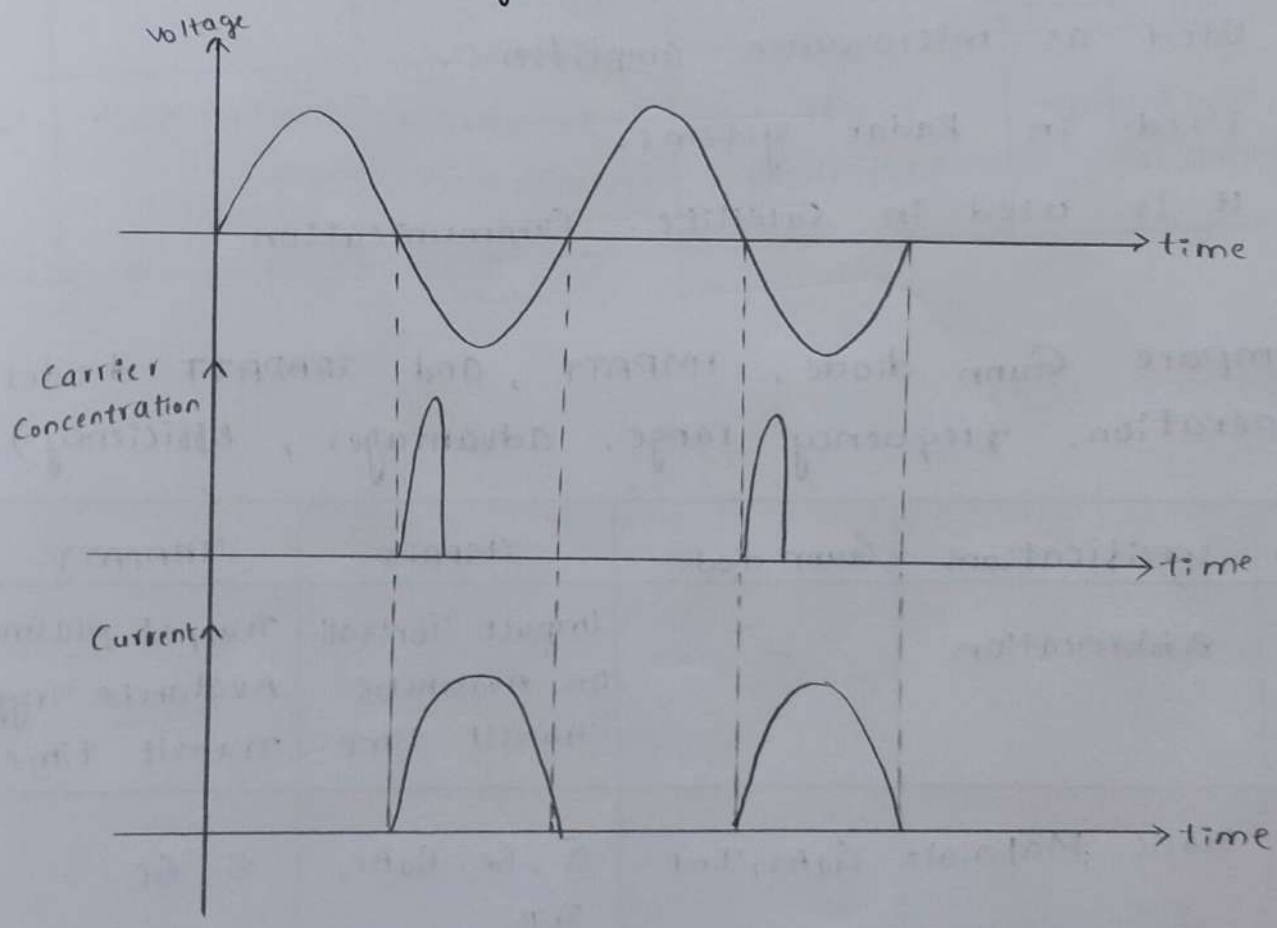
L = Drift region distance

v_s = Saturation velocity

Characteristics :-



With respect to charge carriers :-



Advantages :-

1. Simple to construct & it is of low cost.
2. Small size
3. Operates in negative resistance region.
4. It provides high power capability.

Disadvantages :-

1. High noise levels
2. High operating voltage is required
3. Low Efficiency.

Applications :-

1. Used as microwave oscillators
2. Used as microwave amplifiers.
3. Used in Radar systems.
4. It is used in satellite communication.

4) Compare Gunn diode, IMPATT, and TRAPATT diodes (operation, frequency range, advantages, efficiency).

S.NO	specifications	Gunn diode	IMPATT	TRAPATT.
1.	Abbrevation	-	Impact Ionization Avalanche Transit time	Trapped plasma Avalanche Triggered Transit time.
2.	Basic Materials	GaAs, InP	Si, Ge, GaAs, InP	Si, Ge.

3.	Construction	3 layers $N^+ - N - N^+$	4 - layers $N^+ - P - I - P^+$	3 layers $P^+ - N - N^+$ $N^+ - P - P^+$
4.	Size	small	Small	Small
5.	Biasing	Reverse Biasing	Reverse Biasing	Reverse Biasing
6.	Region of operation	Negative resistance region	Negative resistance region.	Negative resistance region.
7.	Operating freq	1 to 100 GHz	0.5 to 100GHz	1 to 10GHz.
8.	Noise figure	–	High (30dB)	very high (60dB)
9.	Efficiency	–	30% for CW 60% for pulsed waves.	20% to 60% for pulsed waves
10.	Output power	100 to 200W	400W	several hundred watts.
11.	Applications	Microwave Instruments	Amplifiers Oscillators	Amplifiers Oscillators.

5) Describe the procedure for measuring power and frequency of microwaves.

Ans: Measurement of power:

There are 3 different types

1. Low power measurements (0.01mw to 10mw)
   ex: Biometric technique (Bolometer)
2. Medium power measurement (10mw to 1w)
   ex: calorimetric technique (calorimeter)
3. High power measurement (>1watt)
   ex: Calorimetric technique (Calorimeter) (or) (wattmeter)

① Low power measurement:

⇒ Bolometer i.e., Biometric technique can be used.

⇒ 0.01mw to 10mw

⇒ Bolometer device — Positive reflection coefficient
— negative reflection coefficient.

⇒ If temperature increases and resistance increases then it is known as positive temperature coefficient Barrater can be used.

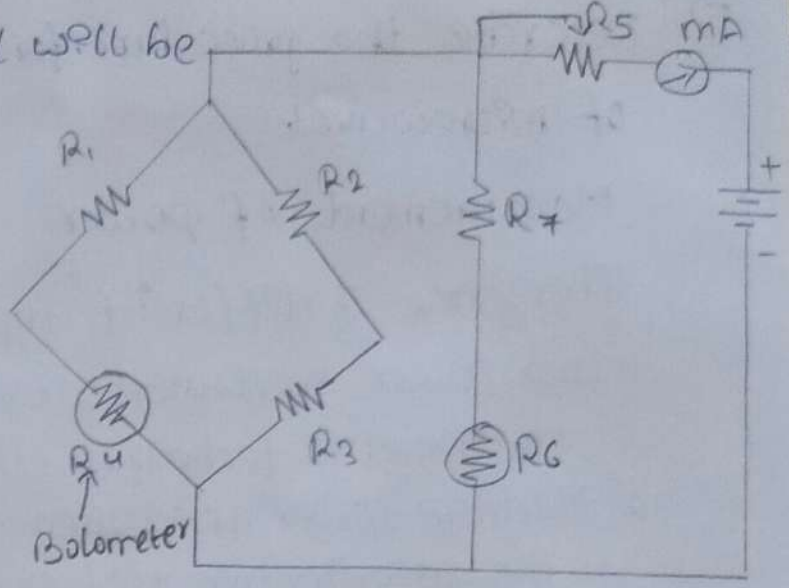
⇒ If temperature increases and resistance decreases then it is known as negative temperature coefficient. Thermister can be used.

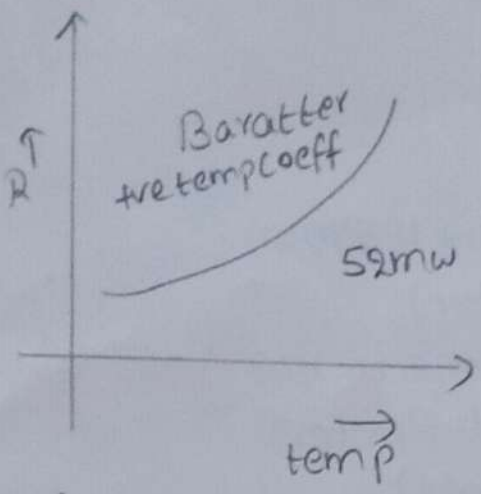
⇒ Bolometer is constructed in bridge structure in which it consists of one arm.

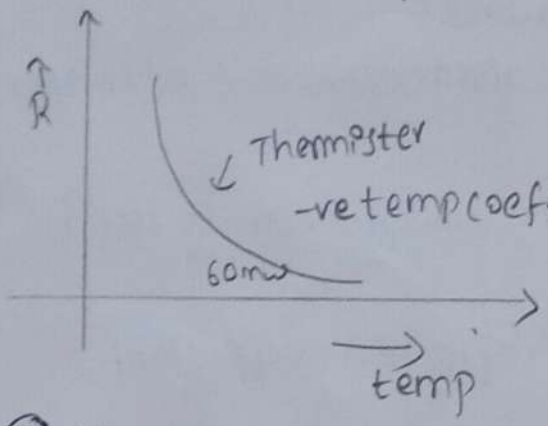
⇒ One arm of the bridge acts as bolometer and it extracts microwave power

⇒ In balancing of the network we make use of adjusting the Vcc.

⇒ Current that is generated will be taken as mA.



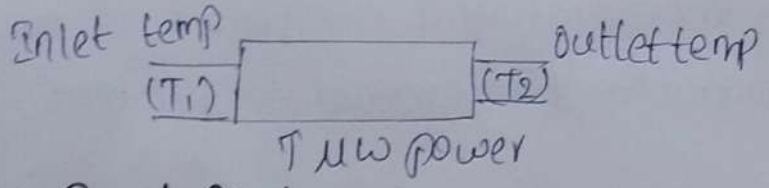




② Medium power measurement:

⇒ Calorimetric technique and 10mw to 1watt

⇒ It relies on temp sensitivity monitoring b/w the inlet (T_1) and outlet (T_2) temp



Calorimetric types:

(a) Static type of calorimeter

we make use of dielectric filled coaxial cables that absorbs heat and measures the power.

(b) Circulating type of calorimeter

we make use of fluid and it absorbs the heat & measures power ex: Thermometer.

③ High power measurement:

⇒ Calorimetric technique and greater than 1 watt.

(a) Dry type

↳ Uses dryelectric medium

(b) Flow type

↳ Uses fluids like water oil.

The fluid changes its temperature when it passes through a load because of absorption in microwave power. Hence the exact power can be calculated as

$$P = \frac{RKe(T_2 - T_1)}{4.18}$$

where,

R = Rate of flow of temperature

K = Specific heat value

e = Specific resistivity

$(T_2 - T_1)$ = Diff B/w outlet and inlet temperature.

6) Describe the procedure for measuring attenuation and impedance of microwaves

Ans: Attenuation: Ratio of input power to the ~~op~~ output power measured in dB.

$$\text{Attenuation (in dB)} = 10\log_{10} P_{in}/P_{out}$$

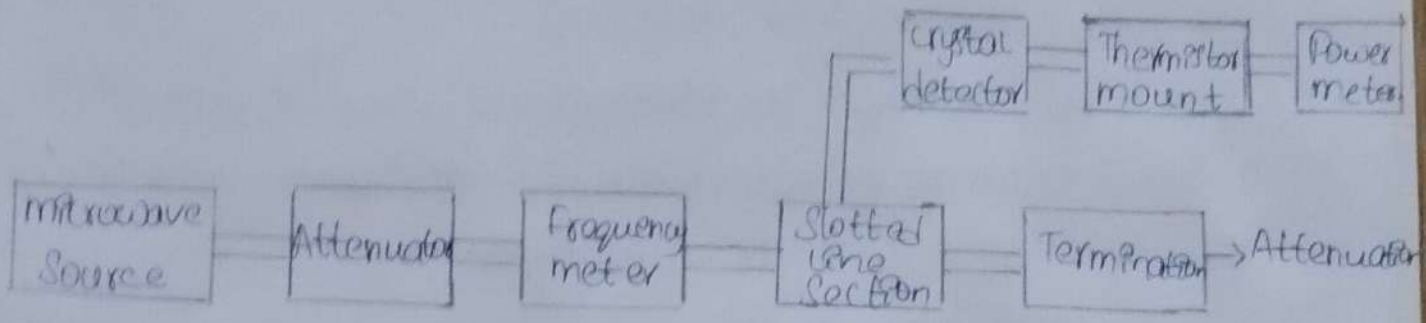
There are 2 diff methods

① Power ratio method:

In this method there are 2 different steps.

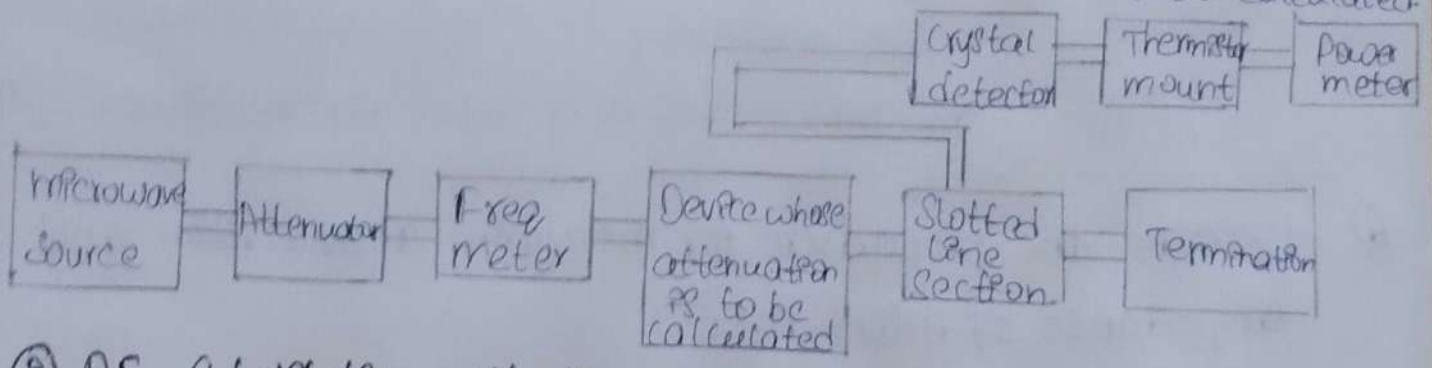
Step-1:

The input and output power of the whole microwave bench is done without the device whose attenuation needs to be calculated.



Step-2:

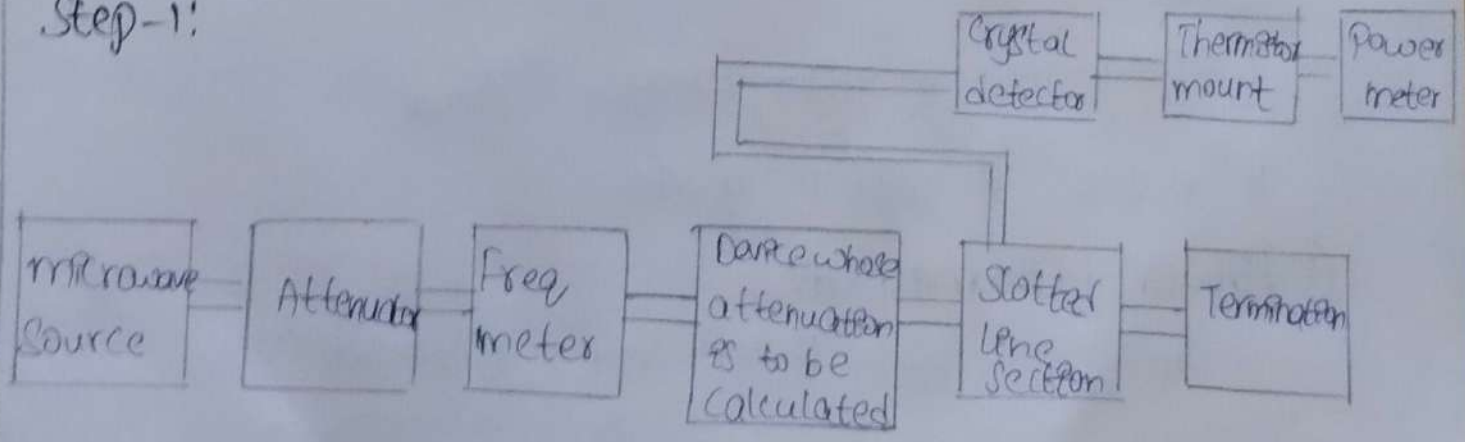
The input and output power of the whole microwave bench is done with the device whose attenuation has to be calculated.



② RF Substitution method:

Accurate of power & attenuation will not obtained in power ratio method.

Step-1:



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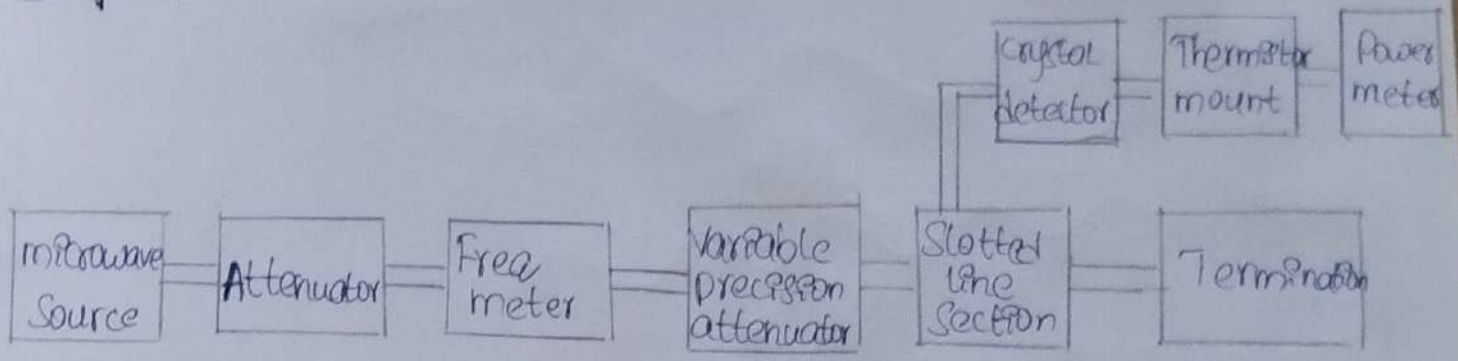


Measurement of impedance

~~Theory~~: There are two diff methods

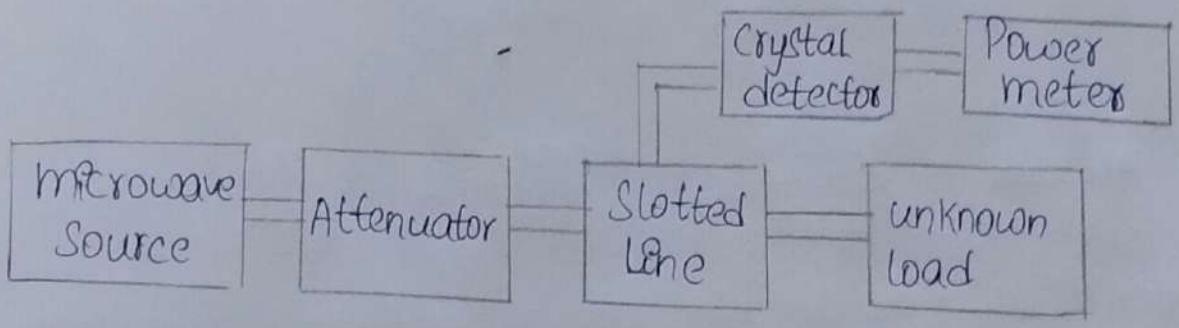
1. Impedance using slotted line
2. Impedance using reflectometer

1. Impedance using slotted line:

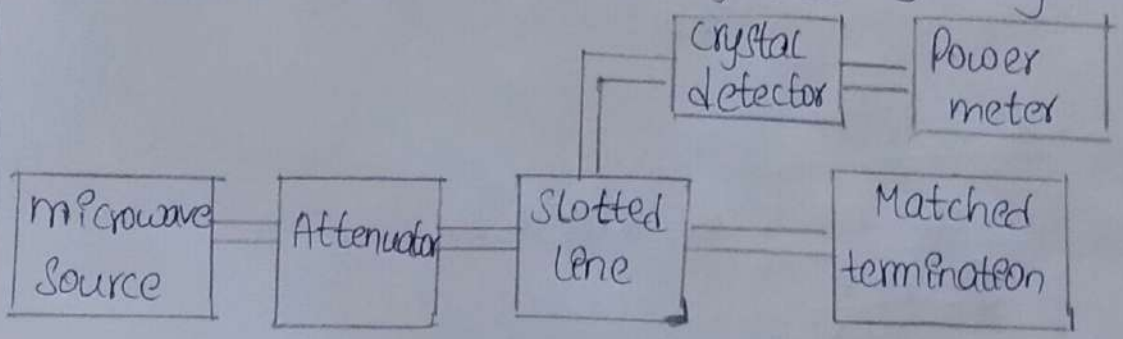
we fix the waveguide to the slotted line for measuring the impedance.

It consists of 2 ways of setups:

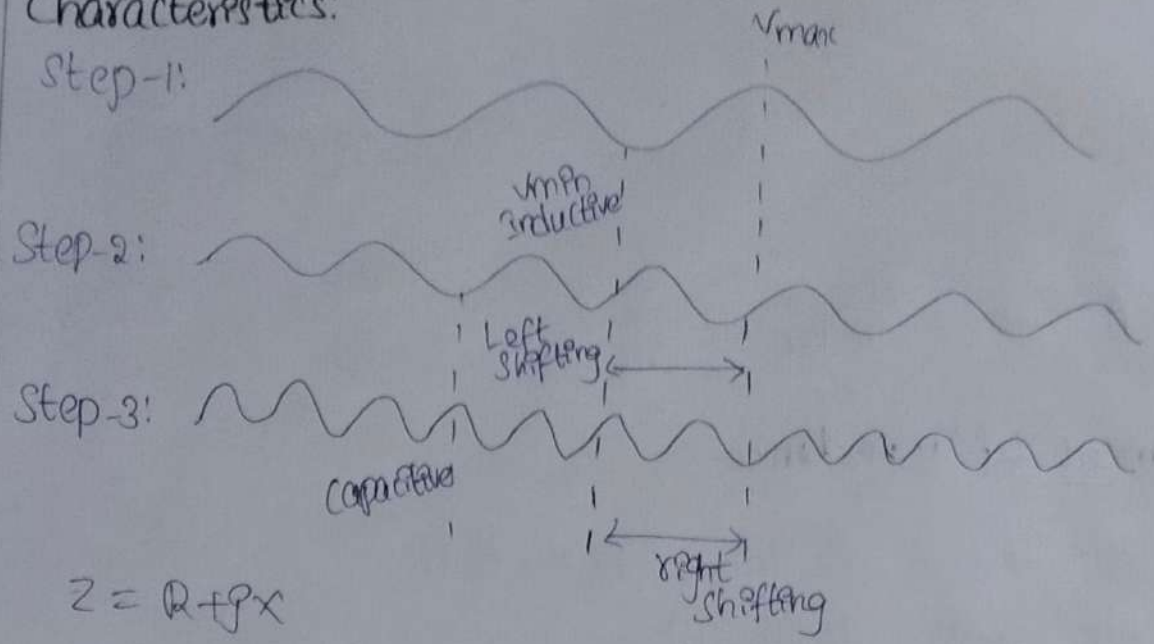
Step-1: Determine the Vmin & Vmax using unknown load.



Step-2: Determine the Vmin & Vmax by using a matched terminat



Characteristics:



$Z = R + jX$

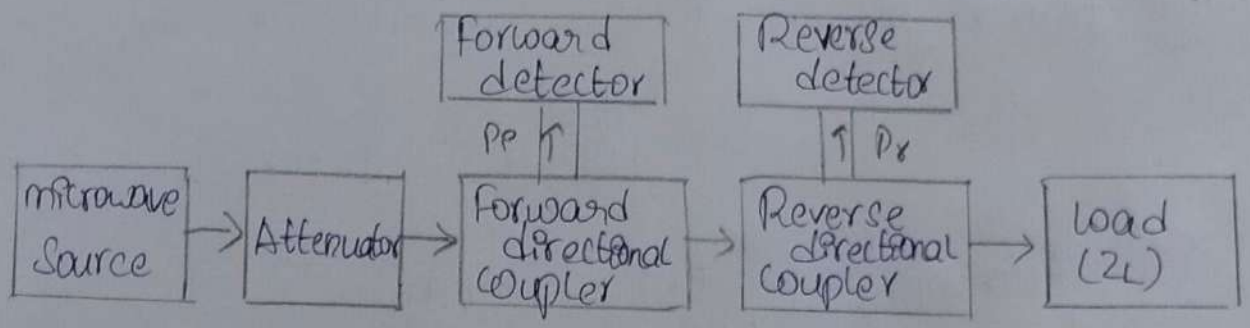
$X > 0 \Rightarrow$ Inductive

$X < 0 \Rightarrow$ Capacitive

2. Impedance using Reflectometer:

⇒ It makes use of 2 directional couplers, place them in opposite direction: One in forward and one in reverse direction.

$$\text{Reflectometer, } \rho = \sqrt{\frac{P_{ref}}{P_{inc}}} = \sqrt{\frac{P_r}{P_i}}$$



Reflectometer is used to obtain the magnitude of reflection coefficient from which the impedance can be calculated.

Reflection coefficient

$$\boxed{\rho = \sqrt{\frac{P_r}{P_i}}}$$

VSWR i.e., 'S' can be calculated as

$$\boxed{S = \frac{1+\rho}{1-\rho}}$$

1) Explain how negative resistance is created in Gunn diodes using two valley theory?

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Symbol :-

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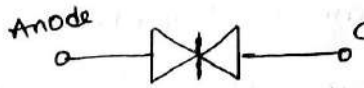

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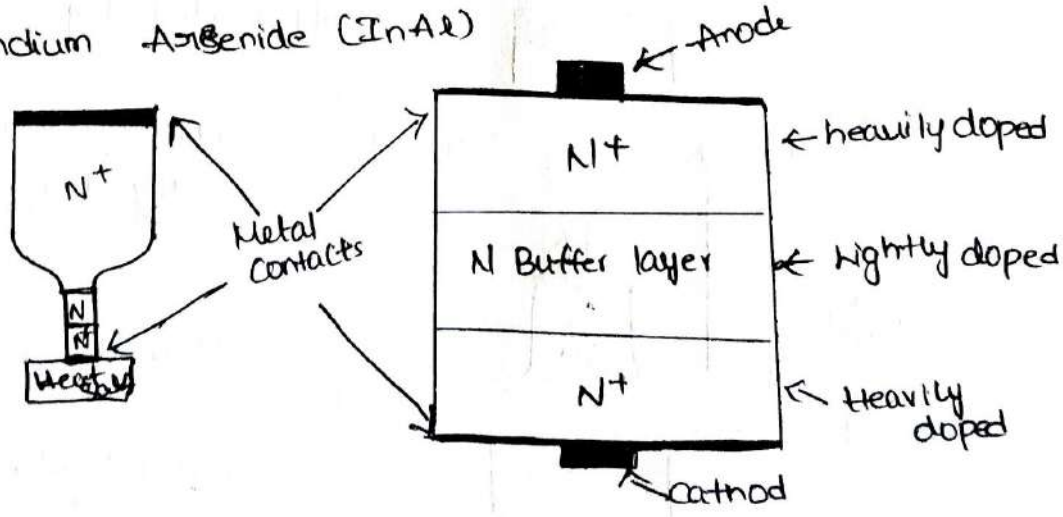
Indium phosphide (Inp)

Indium Arsenide (InAl)



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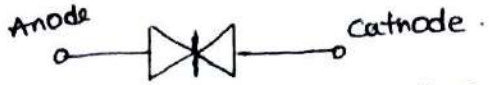

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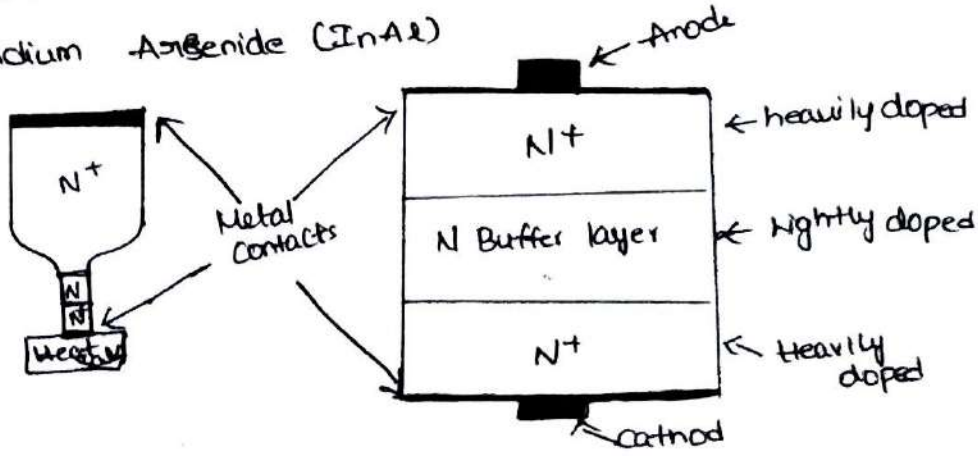

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⇒ While the middle layer (thin layer) is lightly doped an known as buffer layer.

⇒ The width of the middle layer determines the operating frequency of the device.

⇒ The metalic contact for used for providing biasing (input, to the diode).

⇒ Heat sink is connected to prevent over heating of diode.

Working Principle:-

⇒ When voltage is applied an Electric field is Exhibited across the active layer (middle layer).

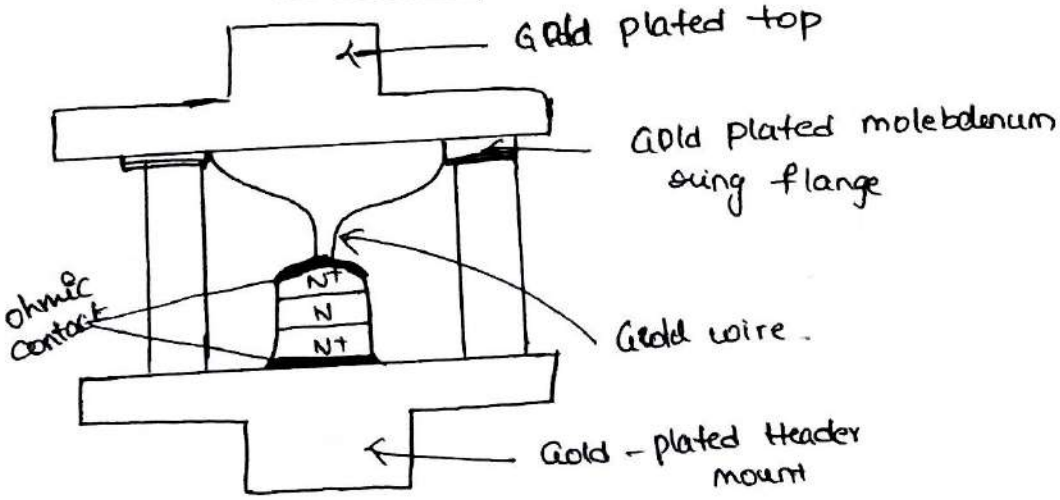
⇒ The current flow increases with the increases in applied voltage.

⇒ Hence the device has +ve resistance.

⇒ The current increase until the voltage increases upto the threshold voltage "Vth".

⇒ When the voltage increases beyond the "Vth". threshold voltage the current density across the active layer decreases.

⇒ This is known as "Gunn Effect"



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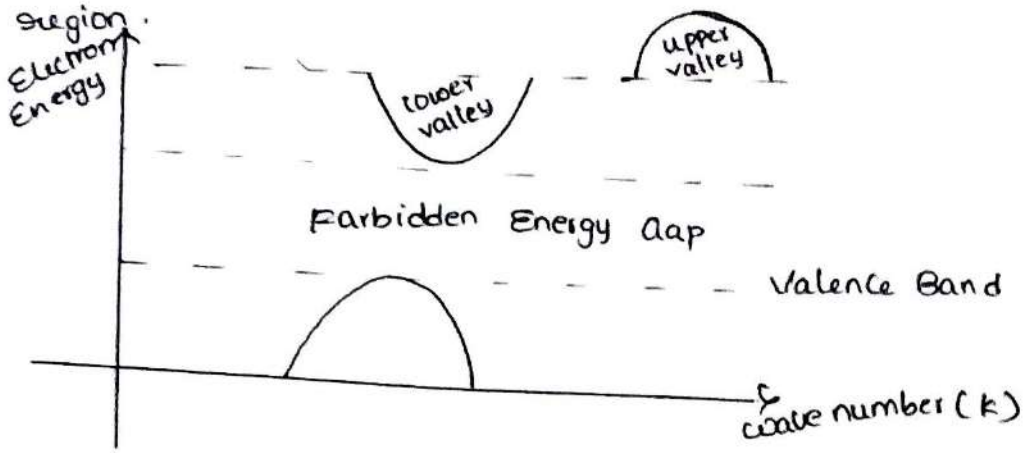
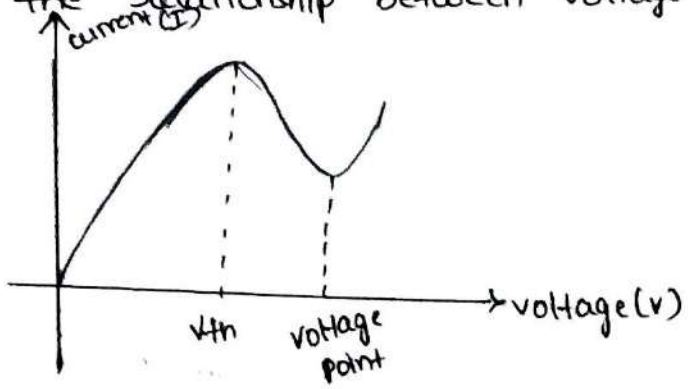
⇒ This doesn't allow the another current pulse to pass through it.

⇒ Hence the operating frequency depends upon the length of the active region.



VI characteristics:-

⇒ It shows the relationship between voltage & current through Gunn diode.



Advantage:-

⇒ It offers very high operating frequency → 10GHz to 200GHz

⇒ It is of small size & occupies little space.

⇒ It has a large operating bandwidth.

Disadvantages:-

⇒ It operates at Negative resistance region at high voltage

⇒ It has low stabality & high Noise figure.

Applications:-

⇒ It is used in microwave TX & RX

⇒ used in Radio communciation devices & sensors.

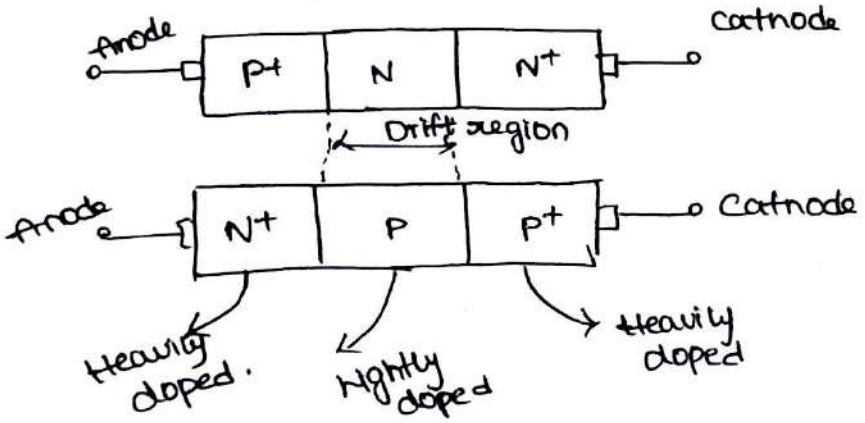
(8) Describe the principle of operation of TRAPATT Diode with suitable diagram.

=> TRAPATT Diode:-

* TRAPATT :- Trapped plasma Avalanche triggered ~~transait~~ transit time diode.
* It is a microwave frequency generator
* The operating frequency range of the diode is from 3GHz-100GHz.
* This diode operates by the creation of plasma in the depletion region.
* plasma has high density charge carries.
* TRAPATT diodes can used as amplifiers oscillators at a frequency range of 0.15 GHz to 100GHz.
* The Efficiency of TRAPATT diodes is more compared to the IMPATT diode.
* The construction of TRAPATT diode is simple & easy to design.

Construction:-

=> It is a 3-layers structure.



⇒ The Extreme Ends of the diode are Heavily doped.

⇒ The middle layer is lightly doped.

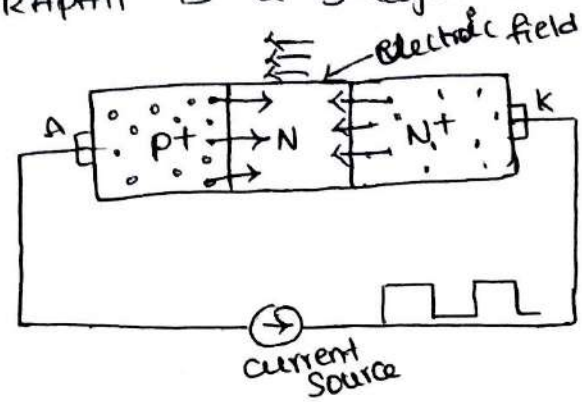
⇒ The middle layer region is called as the drift region.

⇒ The metal contact are used for providing / applying the biasing voltage.

⇒ The two terminals of the diode are anode & cathode.

Working Principle:-

⇒ The TRAPATT is a 3-layer structure diode.



⇒ Initially the diode is connected in reverse biased condition

⇒ The current source is connected to the cathode by which the current passes from N+ to p+ region.

⇒ Now the majority charge carries from N+ region tends to move towards p+ region & vice-versa.

⇒ Due to this diode acts as a capacitor and charge gets increased.

⇒ The increasing points of charge are from A→B.

⇒ As more amount of current gets increased then the electric field in the depletion region gets increased.

⇒ Due to this the capacitor starts slowly discharging

⇒ The discharging points are from B to C.

⇒ As more & more amount of current gets increased the capacitor

discharge more & more.

⇒ So again the discharging points are from to C to D.

⇒ Hence in order to overcome this situation, slowly pull out the electrons & holes from the depletion region.

⇒ Due to this the capacitor changes it's state from discharge ~~the~~ to charging.

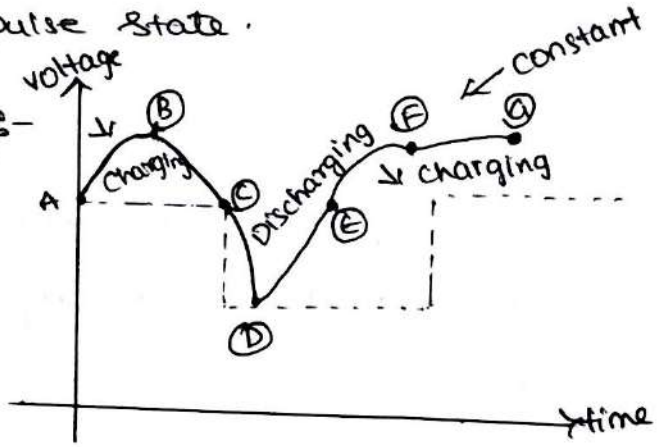
⇒ Now the charging points are from D to E

⇒ Further pulling of charge carries make the capacitor charge more & more.

⇒ So again it charge to the points E to F.

⇒ Finally ~~it's~~ remains constant from F to G points till the next clock pulse state.

Characteristics:-



Advantages:-

* High Efficiency
* Low power dissipation
* Good Frequency range
* High o/p power (1.2kW at 1.5GHz)

Disadvantages:-

* Extremely high noise Figure (60dB)
* unsuitable for continuous wave operation

Applications:-

* Radar System
* Navigational Systems
* Radio Equipments.

3. Which is the principle of operation of IMPATT Diode with Suitable diagram.

A) ⇒ IMPATT - Impact Ionization Avalanche Transit time diode.

⇒ It is a high frequency microwave diode.

⇒ The frequency range is from 1GHZ to 100GHz.

⇒ It provides high power handling Capability.

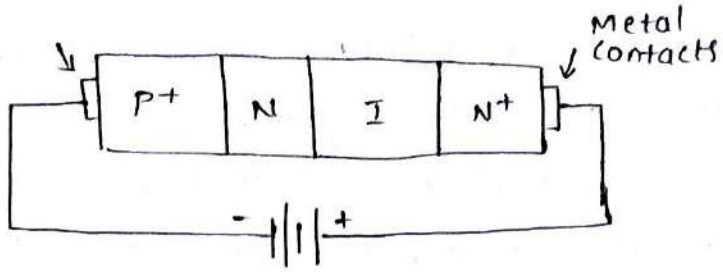
⇒ It is operated at "Negative Resistance region".

⇒ The applying of reverse bias to the diode make it Operate at a breakdown voltage Called as "Avalanche Breakdown voltage".

Construction :-

⇒ It is Constructed by using 4 layers.

⇒ The layers are $P^+ - N - I - N^+$.



⇒ The IMPATT diode can be constructed using different materials like,

* Gallium Arsenide (GaAs)
* Silicon (Si), Germanium (Ge)
* Indium phosphate (Inp).

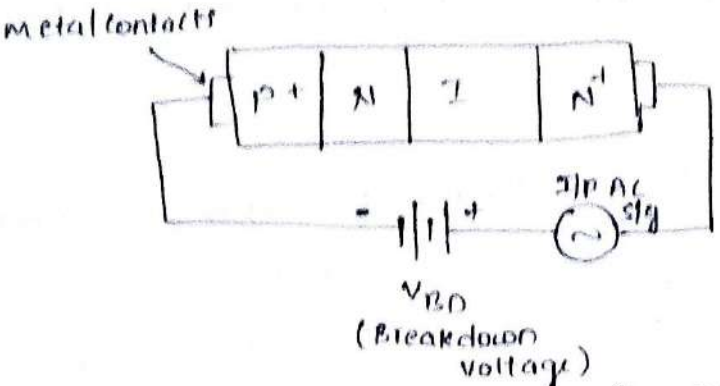
⇒ GaAs is mostly choosen because of low noise behaviour property.

Working principle :-

⇒ It works on mainly two important principles

Avalanche Breakdown

Transit time delay.



⇒ When reverse bias voltage is applied to the diode then the width of the depletion region becomes wider.

⇒ Now the minority charge carriers in both p-type & n-type semiconductor starts moving / crosses over each other.

⇒ Due to this there will be formation of "electron-hole pairs".

⇒ This pairs of electron-hole gets multiplied & more number gets to be generated.

⇒ Hence there will be a release of high amount of current over the depletion region.

⇒ This situation of creation of multiple electron-hole pairs is called as "Avalanche Breakdown (or) Impact Ionization".

Transit Time Delay :-

⇒ The time taken by the charge carriers which are moving from p-type to the N-type semiconductor device is termed as "Transit time delay".

Transit time delay,

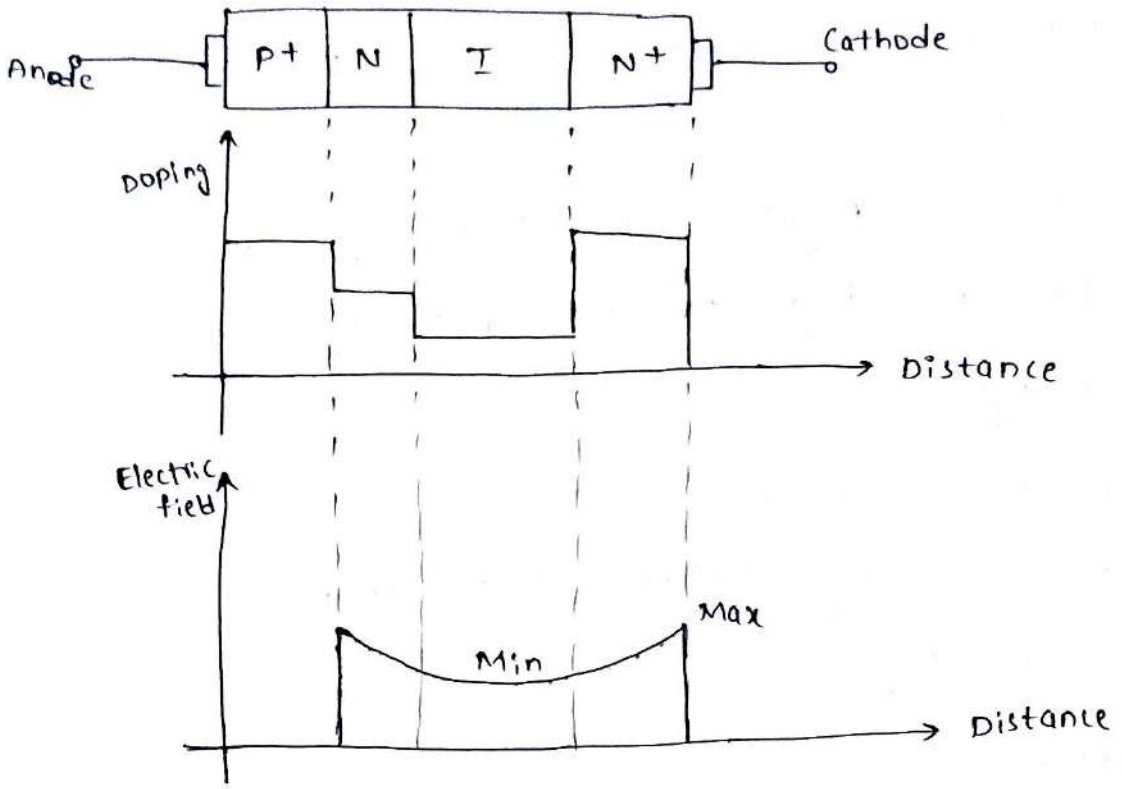
$$\boxed{\tau = L/v_s}$$

Where,

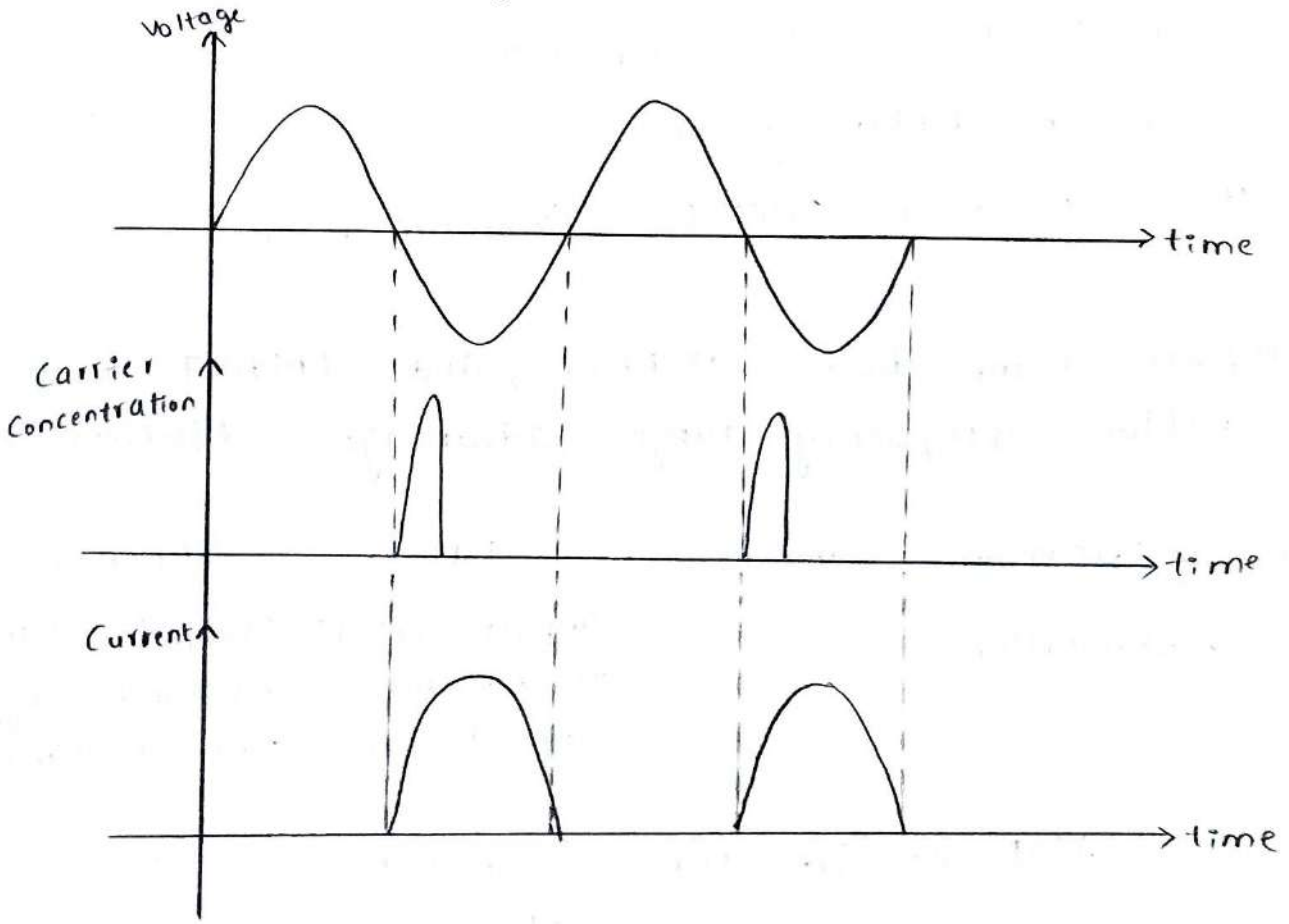
L = Drift region distance

v_s = Saturation velocity

Characteristics :-



With respect to charge carriers :-



Advantages :-

1. Simple to construct & it is low cost.
2. Small size
3. Operates in negative resistance region.
4. It provides high power capability.

Disadvantages :-

1. High noise levels
2. High operating voltage is required
3. Low Efficiency.

Applications :-

1. Used as microwave oscillators
2. Used as microwave amplifiers.
3. Used in Radar systems.
4. It is used in satellite communication.

4) Compare Gunn diode, IMPATT, and TRAPATT diodes (operation, frequency range, advantages, efficiency).

S.NO	specifications	Gunn diode	IMPATT	TRAPATT.
1.	Abbrevation	–	Impact Ionization Avalanche Transit time	Trapped plasma Avalanche Triggered Transit time.
2.	Basic Materials	GaAs, InP	Si, Ge, GaAs, InP	Si, Ge.

3.	Construction	3 layers $N^+ - N - N^+$	4 - layers $N^+ - P - I - P^+$	3 layers $P^+ - N - N^+$ $N^+ - P - P^+$
4.	Size	Small	Small	Small
5.	Biasing	Reverse Biasing	Reverse Biasing	Reverse Biasing
6.	Region of operation	Negative resistance region	Negative resistance region.	Negative resistance region.
7.	Operating freq	1 to 100 GHZ	0.5 to 100GHZ	1 to 10GHZ.
8.	Noise figure	–	High (30dB)	very high (60dB)
9.	Efficiency	–	30% for CW 60% for pulsed waves.	20% to 60% for pulsed waves
10.	Output power	100 to 200W	400W	several hundred watts.
11.	Applications	Microwave Instruments	Amplifiers Oscillators	Amplifiers Oscillators.

5) Describe the procedure for measuring power and frequency of microwaves.

Ans: Measurement of power:

There are 3 different types

1. Low power measurements (0.01mw to 10mw)
   ex: Biometric technique (Bolometer)
2. Medium power measurement (10mw to 1w)
   ex: calorimetric technique (calorimeter)
3. High power measurement (>1watt)
   ex: Calorimetric technique (Calorimeter) (or) (wattmeter)

(1) Low power measurement:

=> Bolometer i.e., Biometric technique can be used.

=> 0.01 mw to 10mw

=> Bolometer device — Positive reflection coefficient
                — negative reflection coefficient.

=> If temperature increases and resistance increases then it is known as positive temperature coefficient Barrater can be used.

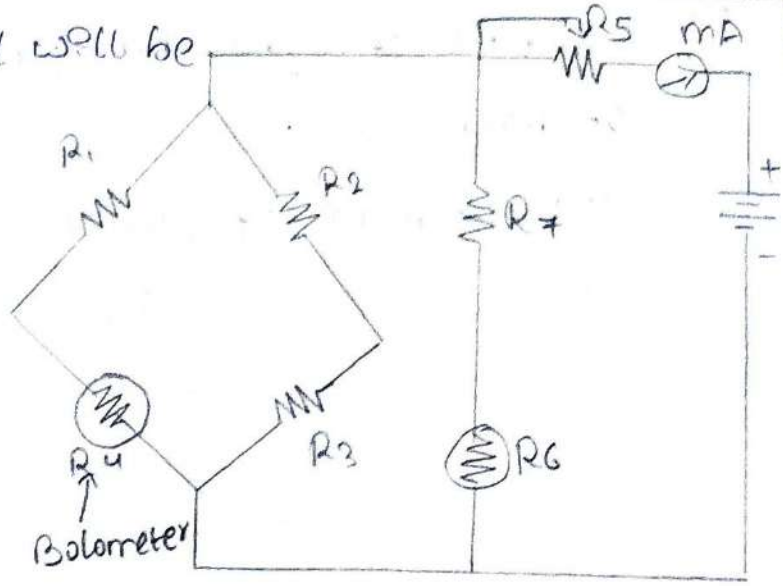
=> If temperature increases and resistance decreases then it is known as negative temperature coefficient. Thermister can be used.

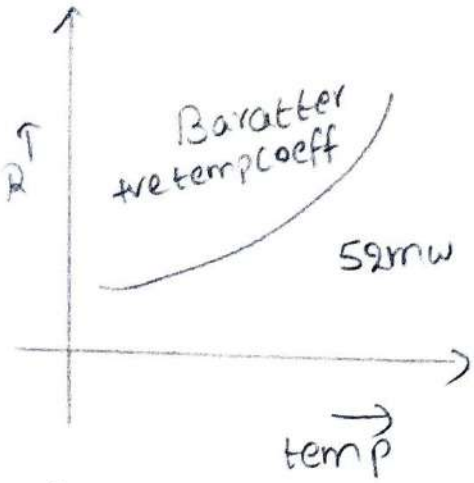
=> Bolometer is constructed in bridge structure in which it consists of one arm.

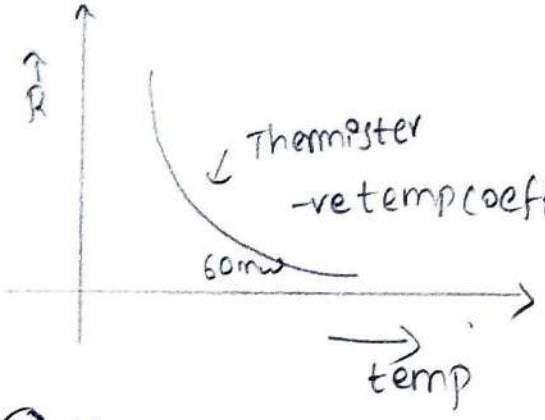
=> One arm of the bridge acts as bolometer and it extracts microwave power

=> In balancing of the network we make use of adjusting the Vcc.

⇒ Current that is generated will be taken as mA.



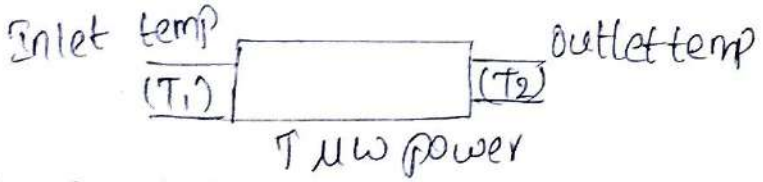




② Medium power measurement:

⇒ Calorimetric technique and 10mw to 1watt

⇒ It relies on temp sensitivity monitoring b/w the inlet (T_1) and outlet (T_2) temp



Calorimetric types:

(a) Static type of calorimeter

we make use of dielectric filled coaxial cables that absorbs heat and measures the power.

(b) Circulating type of calorimeter

we make use of fluid and it absorbs the heat & measures power ex: Thermometer.

③ High power measurement:

⇒ Calorimetric technique and greater than 1 watt

(a) Dry type

↳ uses dryelectric medium

(b) Flow type

↳ uses fluids like water oil.

The fluid changes its temperature when it passes through a load because of absorption in microwave power Hence the exact power can be calculated as

$$P = \frac{RKe(T_2 - T_1)}{4.18}$$

where,

R = Rate of flow of temperature

K = Specific heat value

e = Specific resistivity

$(T_2 - T_1)$ = Diff B/w outlet and inlet temperature.

6) Describe the procedure for measuring attenuation and impedance of microwaves

Ans: Attenuation: Ratio of input power to the ~~op~~ output power measured in dB.

$$\text{Attenuation (in dB)} = 10\log_{10} P_{in}/P_{out}$$

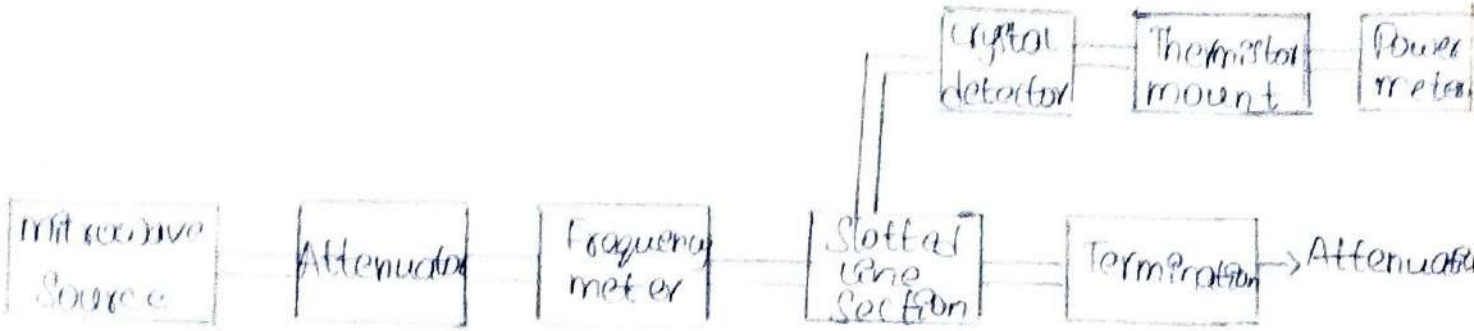
There are 2 diff methods

① Power ratio method:

In this method there are 2 different steps.

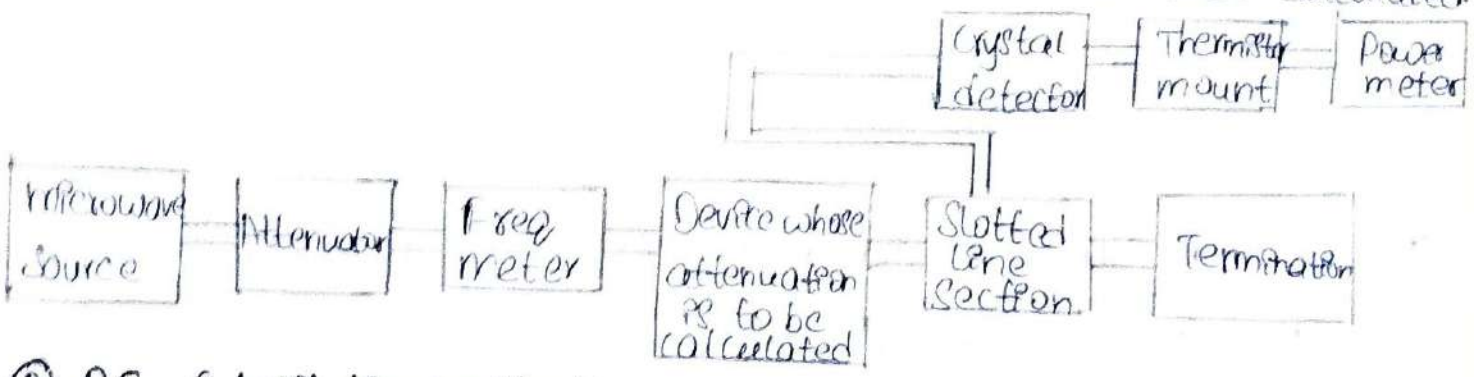
Step 1:

The input and output power of the whole microwave bench is done without the device whose attenuation needs to be calculated.



Step 2:

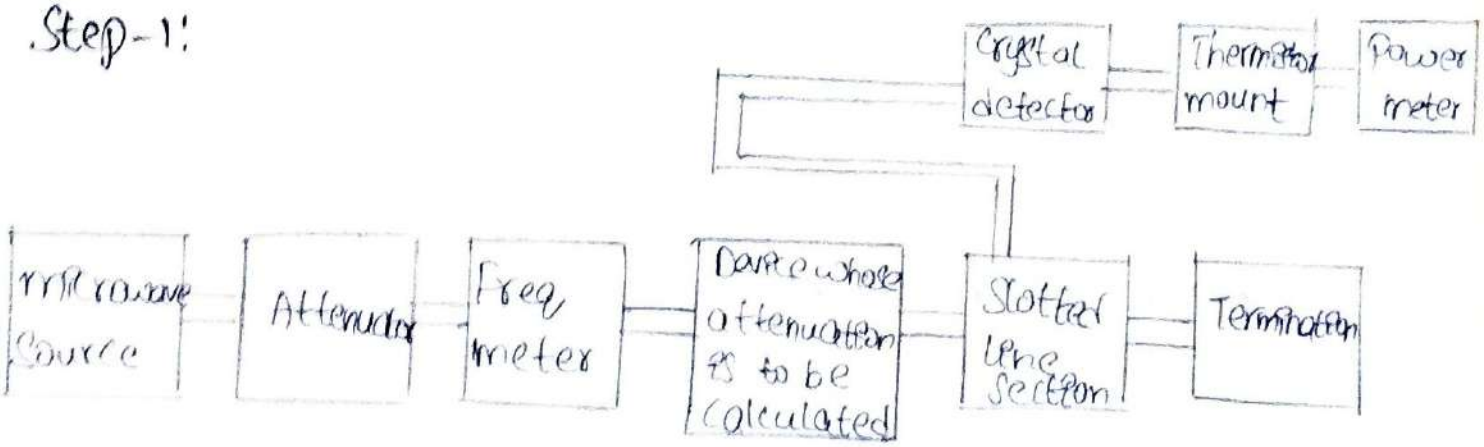
The input and output power of the whole microwave bench is done with the device whose attenuation has to be calculated.



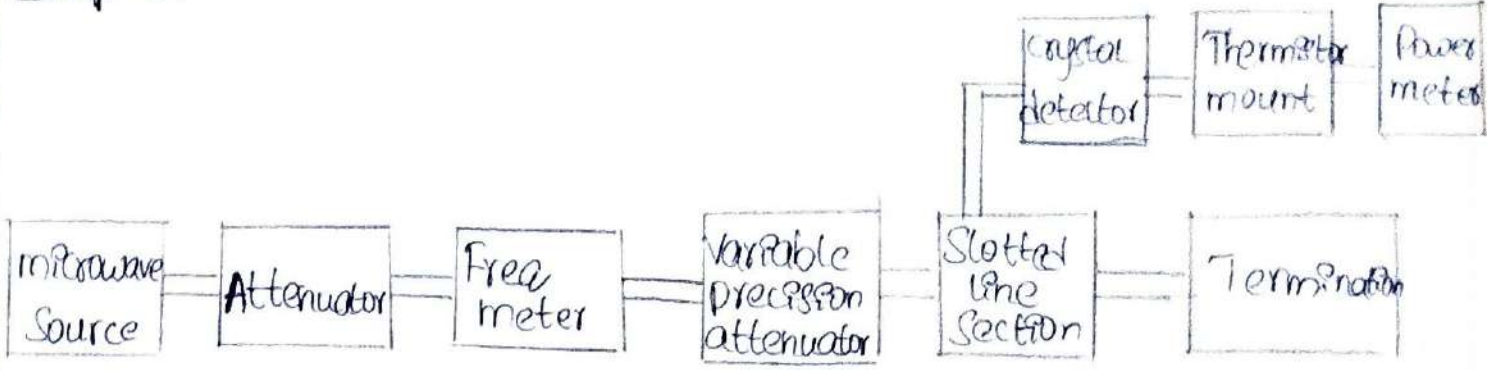
② RF Substitution method:

Accurate of power & attenuation will not obtained in power ratio method.

Step-1:



Step-2:



Measurement of Impedance

~~Theory~~ There are two diff methods

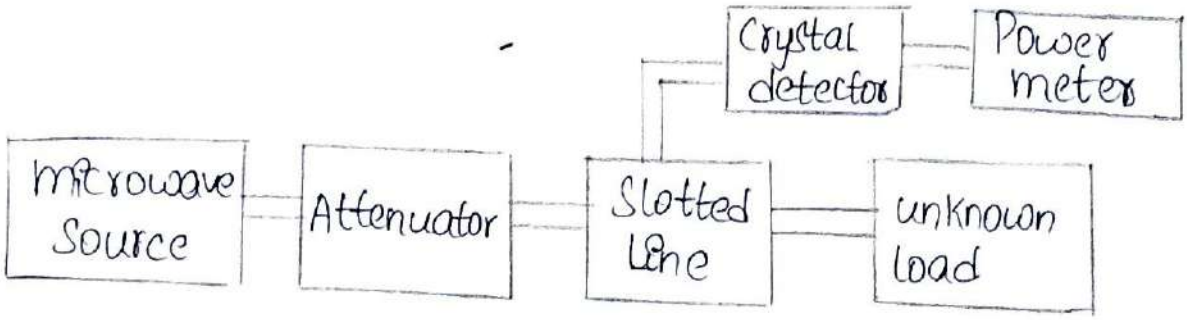
1. Impedance using slotted line
2. Impedance using reflectometer

1. Impedance using slotted line:

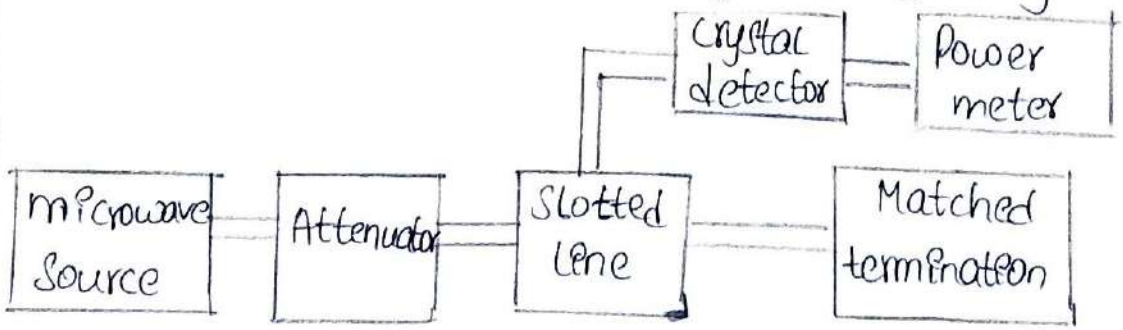
we fix the waveguide to the slotted line for measuring the impedance.

It consists of 2 ways of setups:

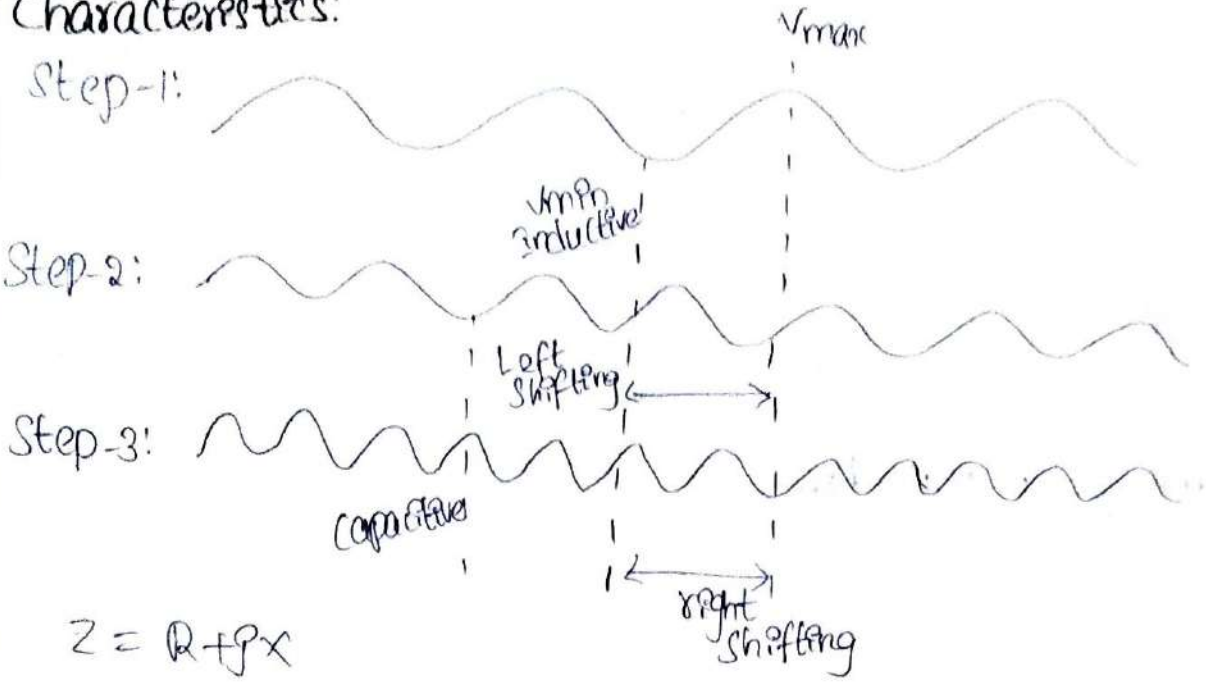
Step-1: Determine the Vmin & Vmax using unknown load.



Step-2: Determine the Vmin & Vmax by using a matched termina



Characteristics:



$Z = R + jX$

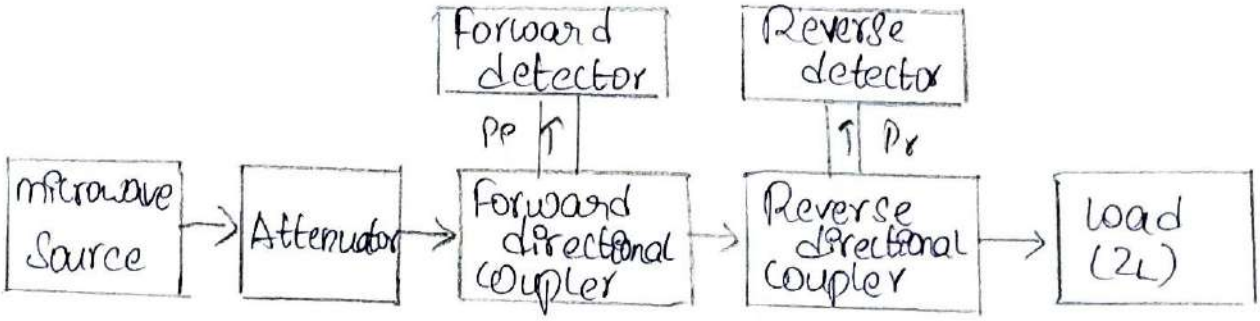
$X > 0 \Rightarrow$ Inductive

$X < 0 \Rightarrow$ Capacitive

2. Impedance using Reflectometer:

⇒ It makes use of 2 directional couplers, place them in opposite direction: One in forward and one in reverse direction

$$\text{Reflectometer, } \rho = \sqrt{\frac{P_{ref}}{P_{inc}}} = \sqrt{\frac{P_r}{P_i}}$$



Reflectometer is used to obtain the magnitude of reflection coefficient from which the impedance can be calculated.

Reflection coefficient

$$\boxed{\rho = \sqrt{\frac{P_r}{P_i}}}$$

VSWR i.e., 'S' can be calculated as

$$\boxed{S = \frac{1+\rho}{1-\rho}}$$